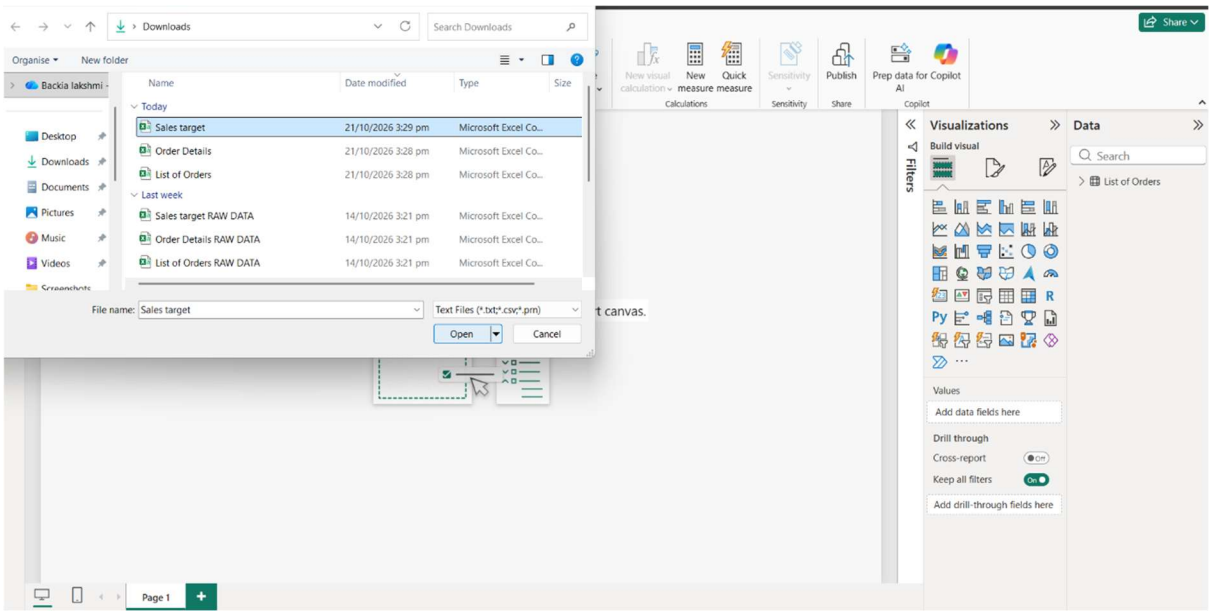
Dax and Data Visualization

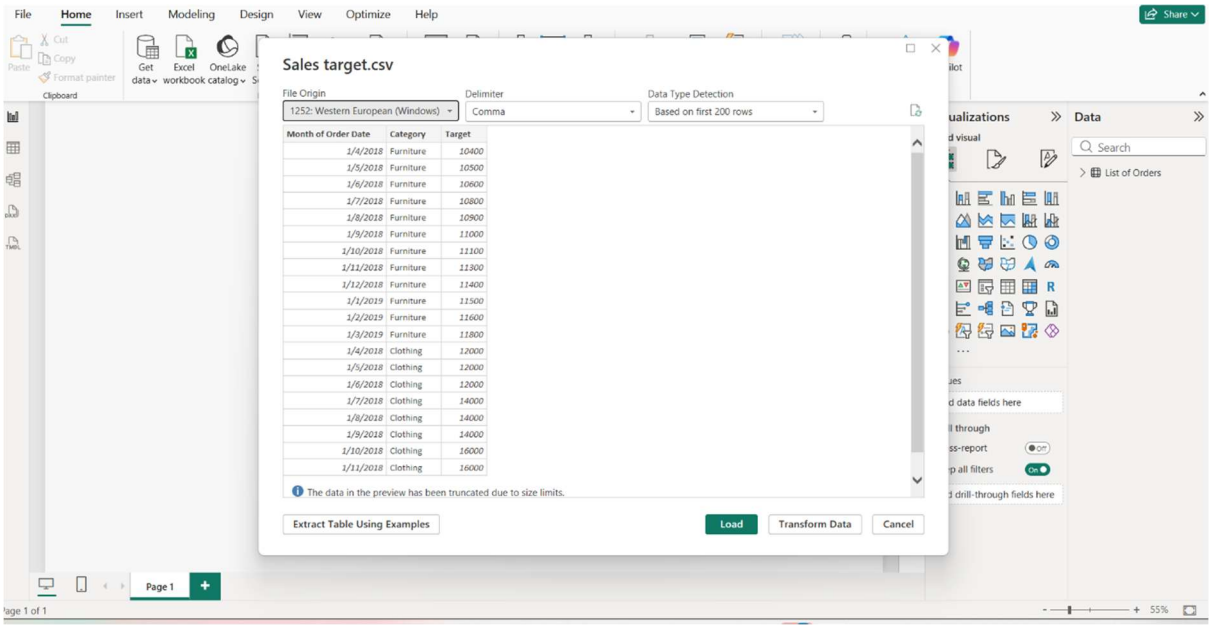
Project description

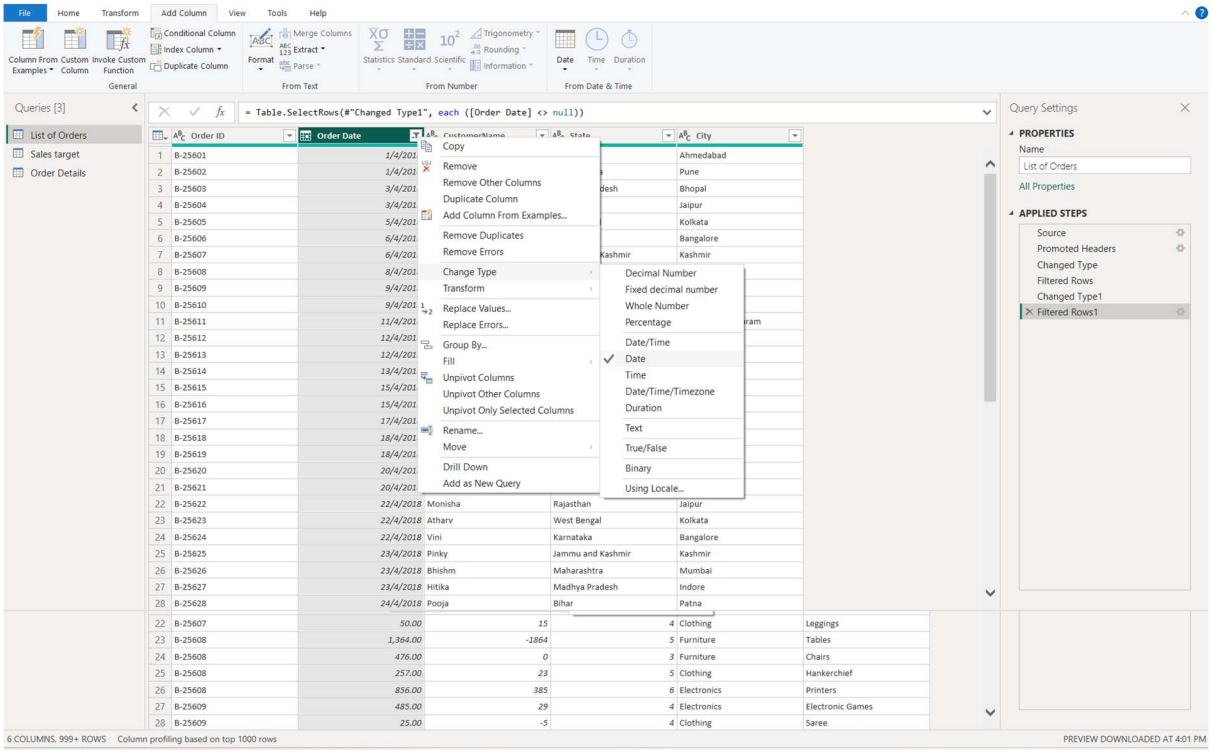
This project focuses on analyzing retail sales performance using Microsoft Power BI. The dataset contains order details, customer and geographical information, product categories, sales amounts, profits, quantities, and monthly sales targets.The project uses Power Query for data cleaning, DAX (Data Analysis Expressions) for calculating key performance indicators (KPIs), and Power BI visualizations to identify sales trends, profitability, category performance, and target achievement.

COLUMN	DESCRIPTION
Order ID	Unique identifier assigned to an order.
Amount	Monetary value of the sales transaction.
Profit	Profit or loss earned from the transaction.
Quantity	Number of units sold in the transaction.
Category	Main product category, such as Furniture, Clothing, or Electronics.
Sub-Category	More specific classification of a product within its main category.
Month of Order Date	Month and year for which the sales target is defined.
Category	Product category for which the target is set.
Target	Planned sales amount for the specified month and category.
Location	Cutomers easily identified.

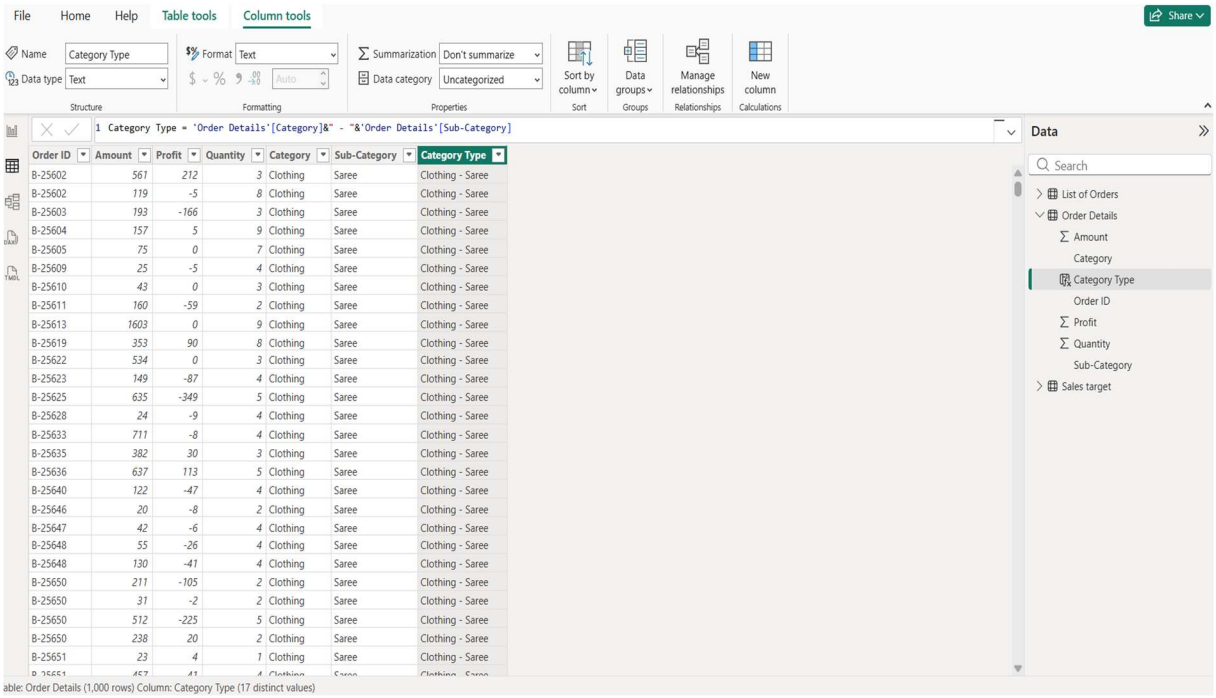
- Import List of Orders.csv into Power BI.



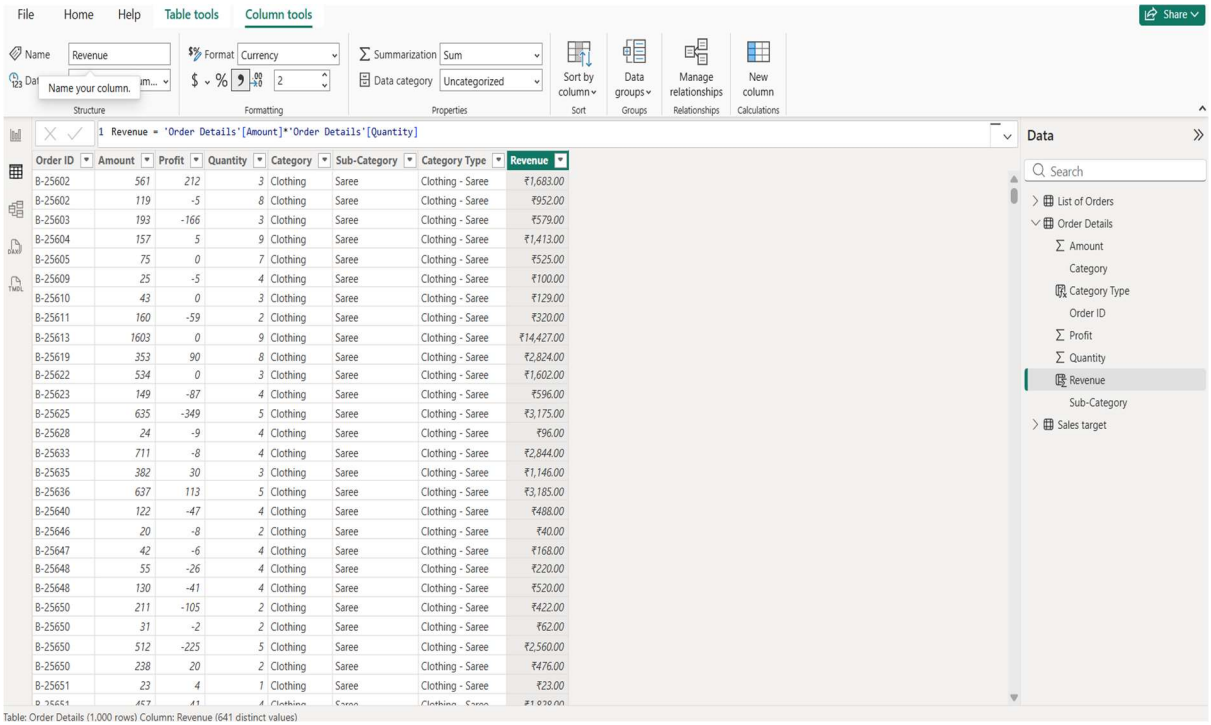


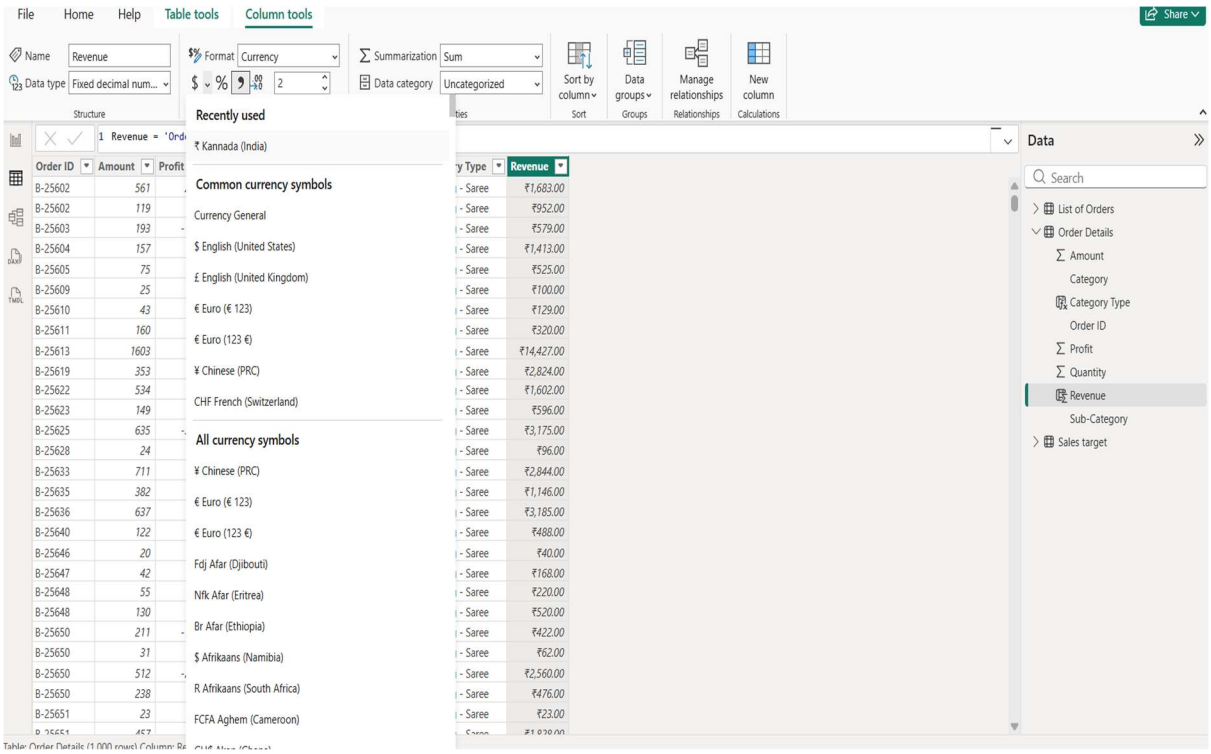


Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

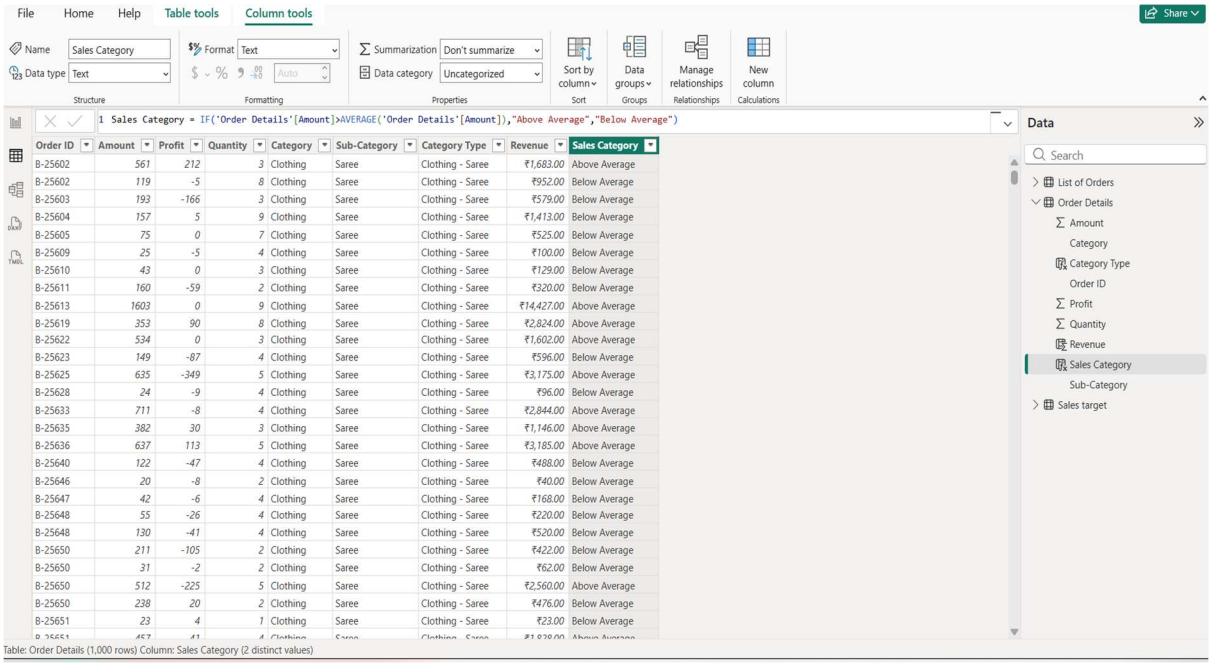


- **Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.**
- **Revenue = `'Order Details'[Amount]*'Order Details'[Quantity]`**

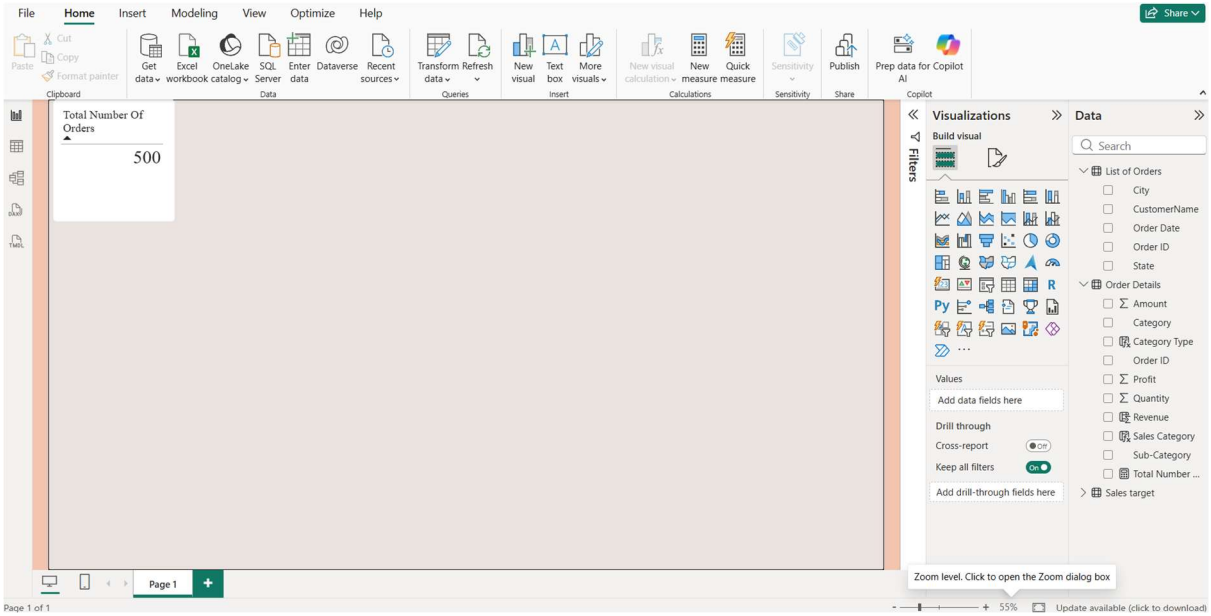




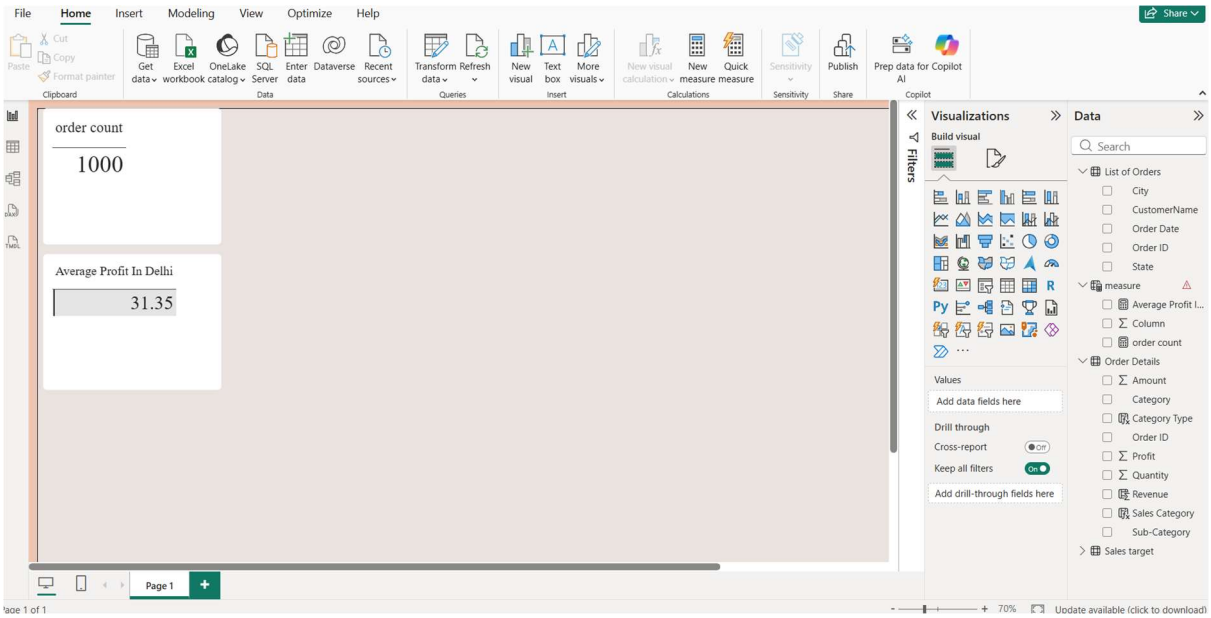
- **Add a calculated column named ‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.**
- **`Sales Category = IF('Order Details'[Amount]>AVERAGE('Order Details'[Amount]),"Above Average","Below Average")`**



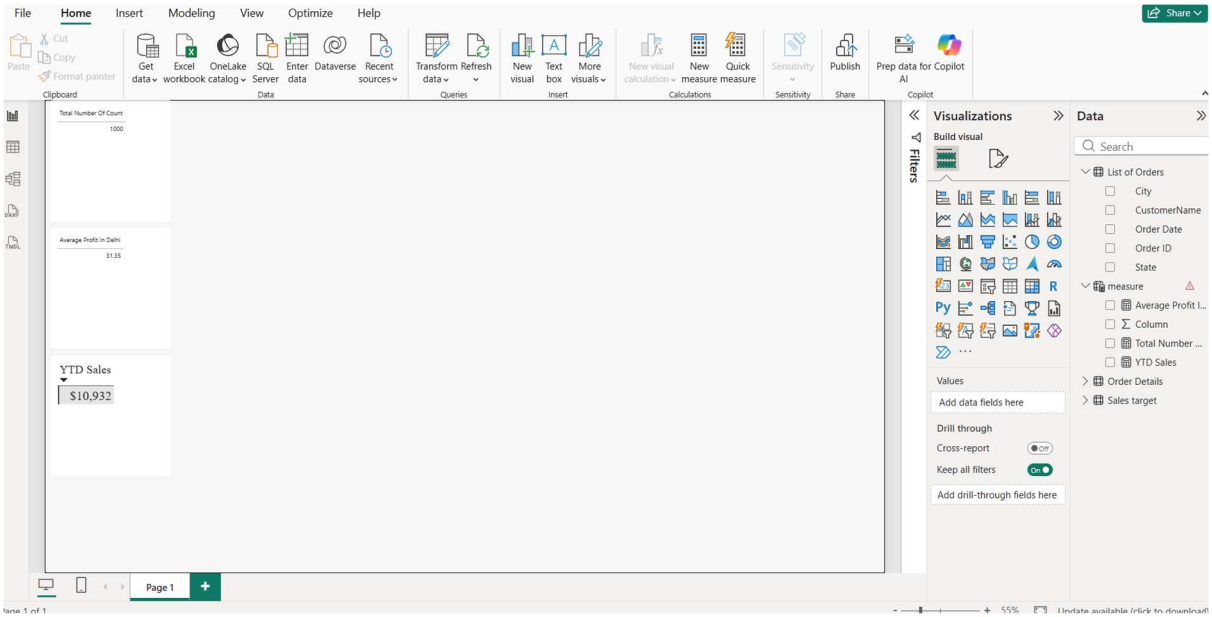
- **Create a measure that calculates the total number of orders in the Order Details table.**
- **`Total Number Of Orders = COUNT('Order Details'[Order ID])`**



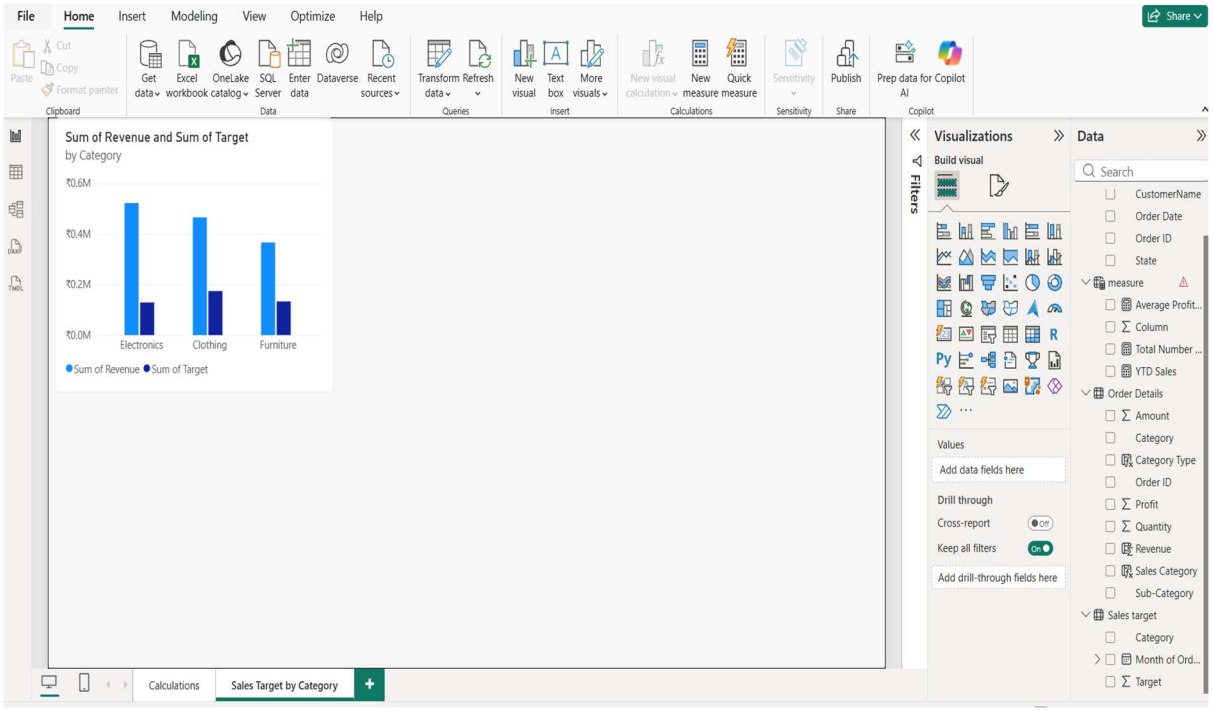
- **Create a measure that calculates the average profit for orders placed in Delhi.**
- **`Average Profit In Delhi = CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[City]="Delhi")`**



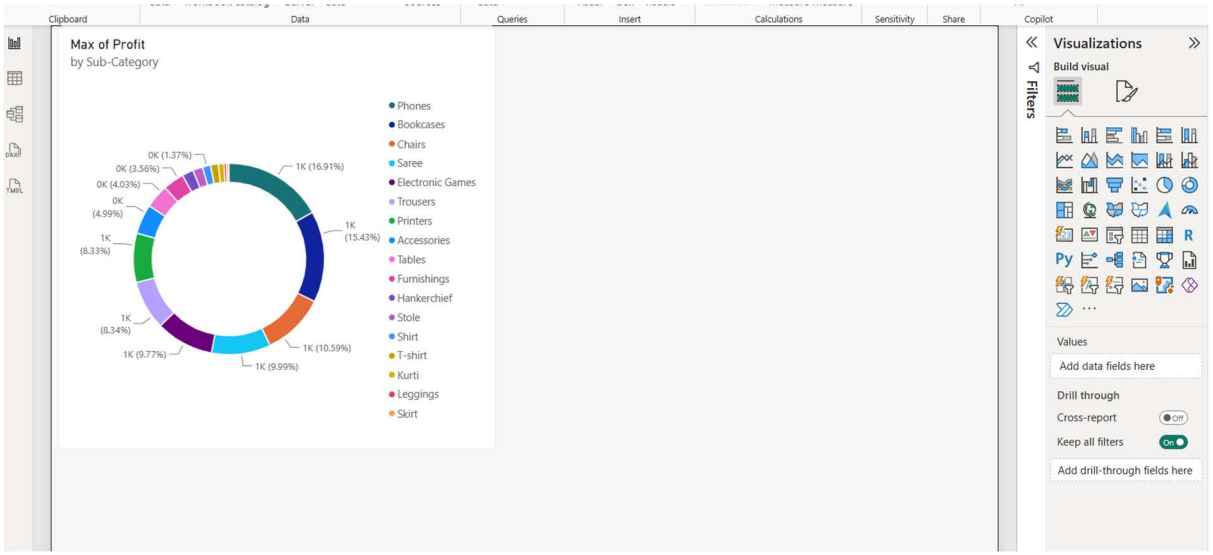
- **Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.**
- **`YTD Sales = TOTALYTD(SUM('Order Details'[Amount]),'List of Orders'[Order Date])`**



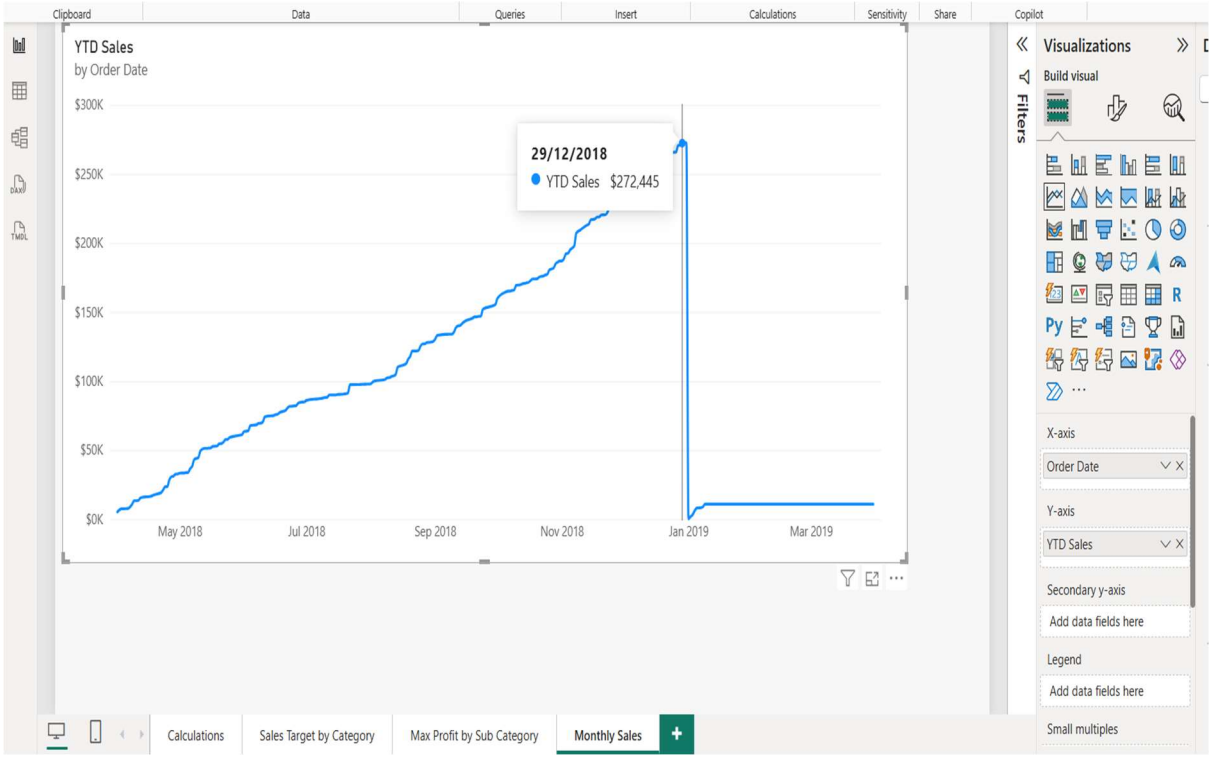
- **Compare actual sales with sales targets by category using a clustered column chart.**



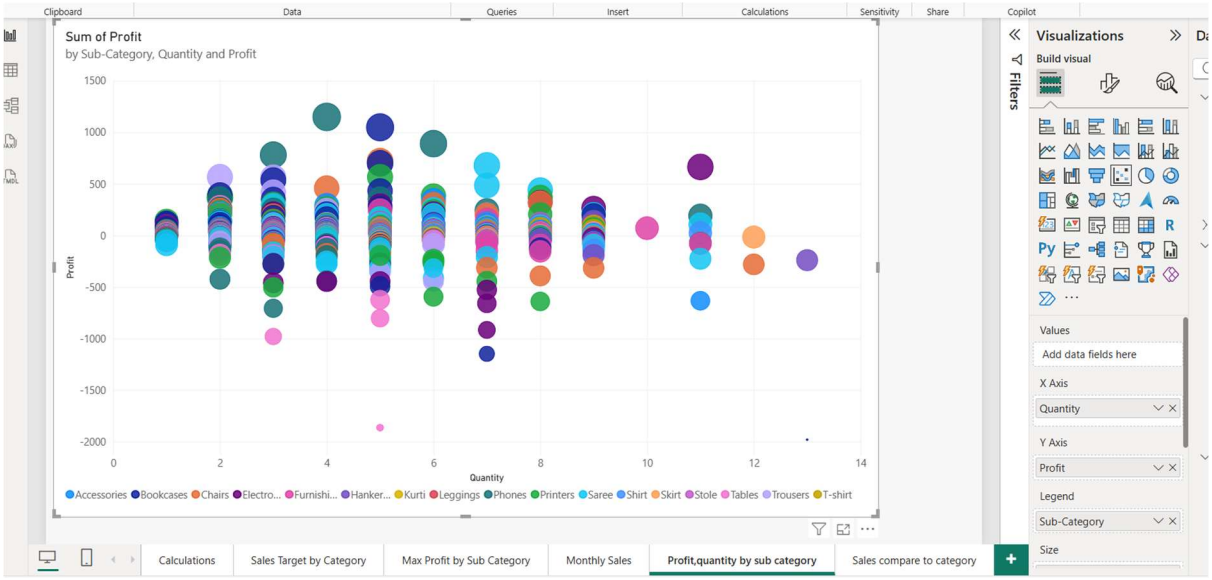
- **Analyze the maximum profit margin for each sub-category of products using a donut chart.**



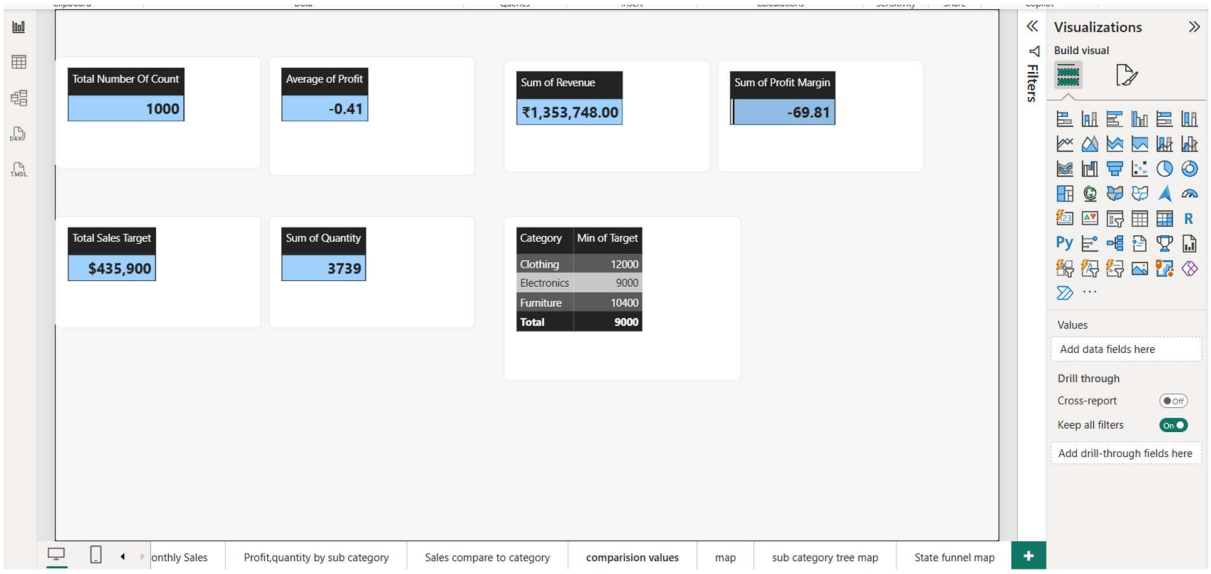
- **The trend of monthly sales over time using a line chart.**



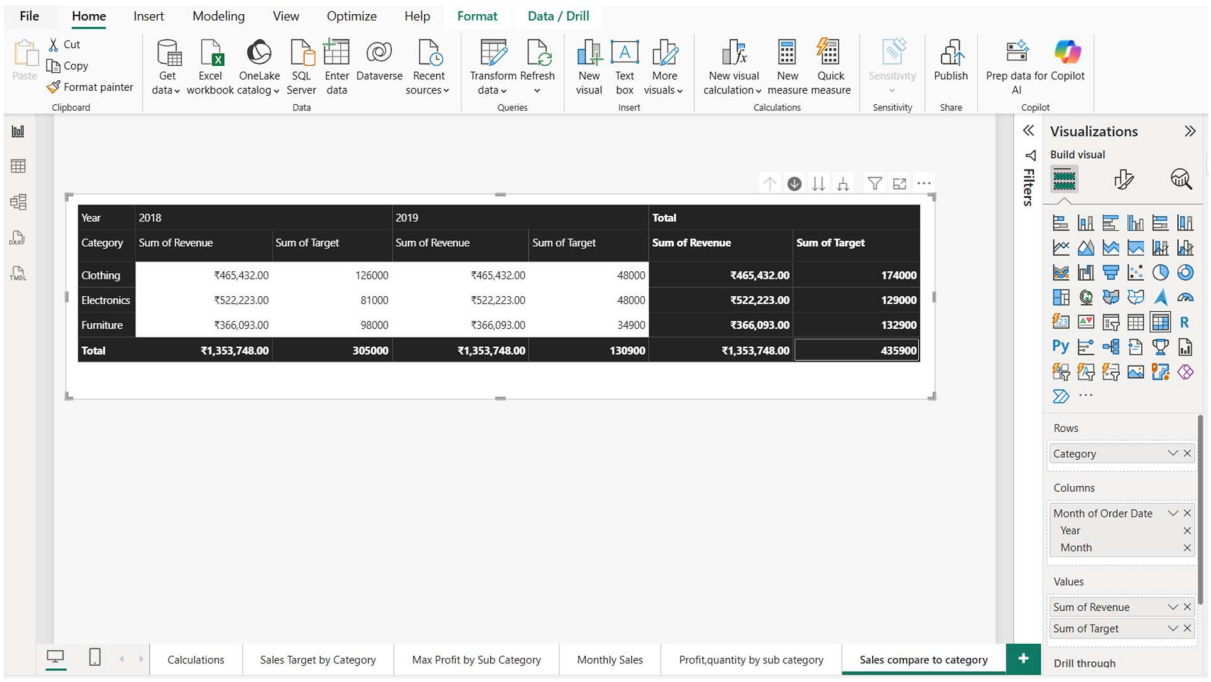
- **Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.**



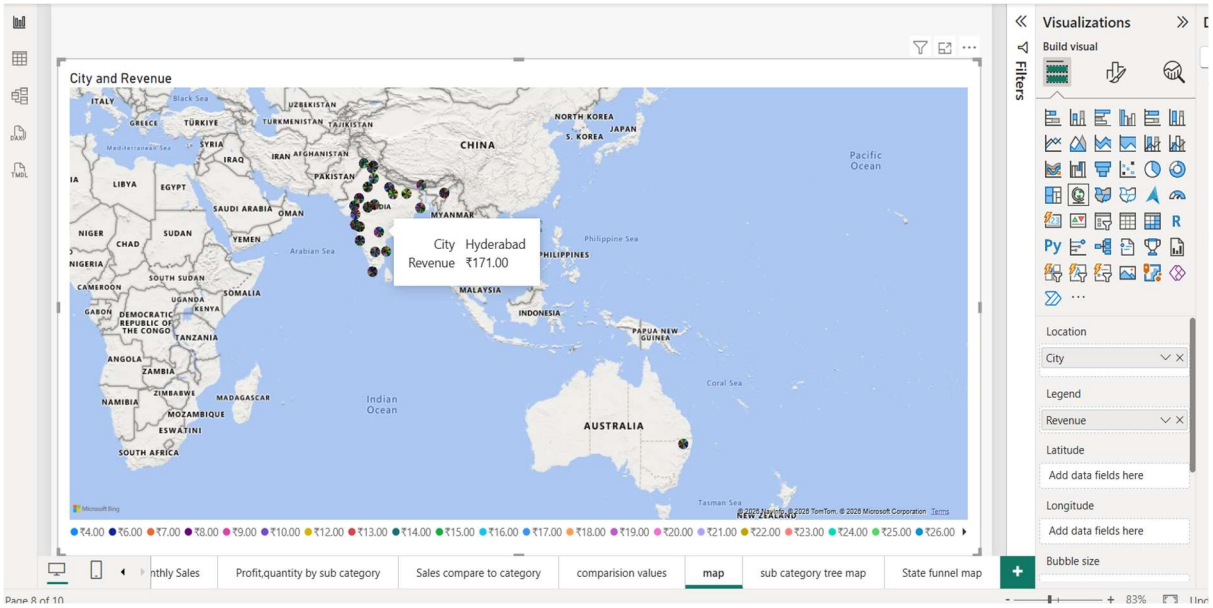
- **Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.**



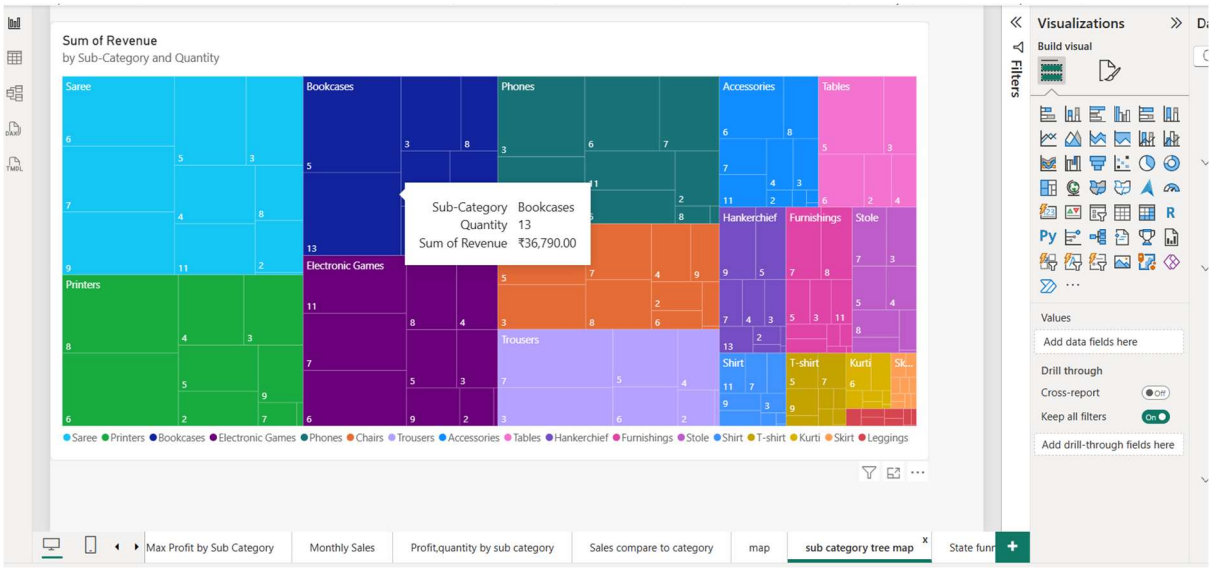
- **Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.**



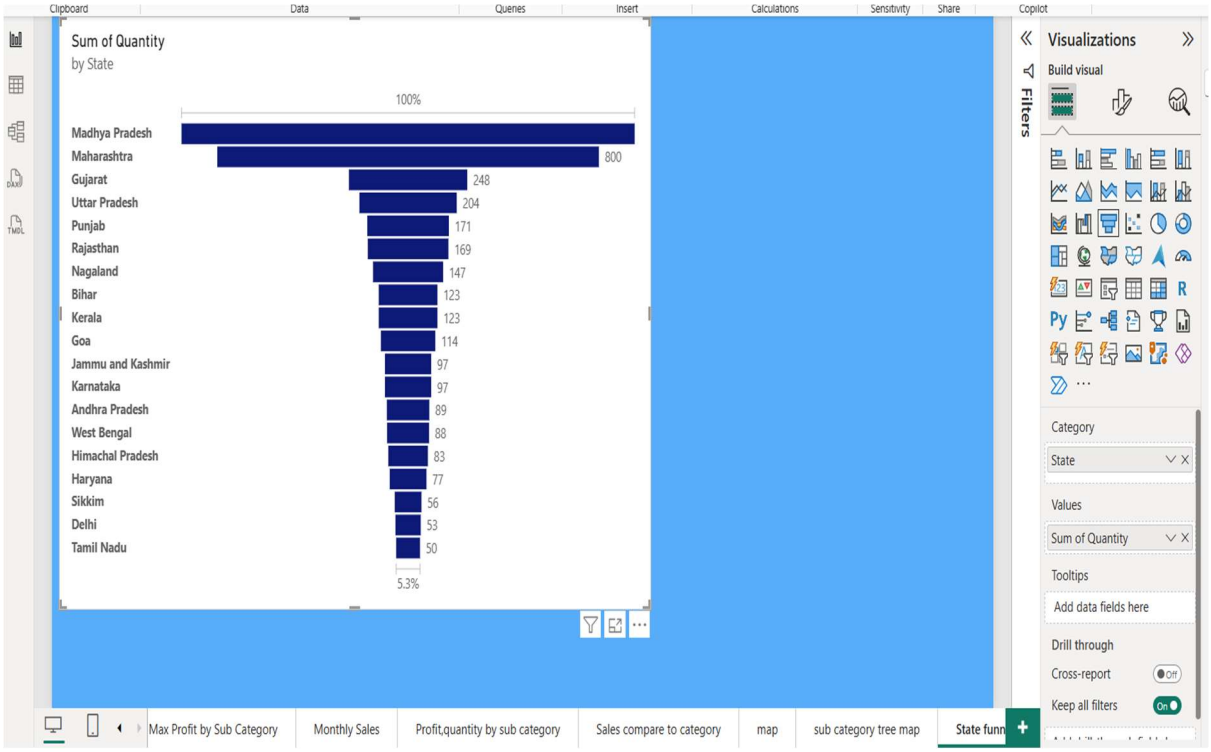
Visualize total sales on a map by city to identify regional sales patterns.



Represent the sales distribution across Different sub-categories using a treemap.



- **Create a funnel chart to visualize the distribution of order counts across different states.**



Conclusion :

This Sales Performance Analysis project demonstrates how Power BI, DAX, and data visualization can transform raw retail data into meaningful business insights. The analysis measures sales, profit, quantity, profit margins, and sales target achievement while comparing performance across categories, sub-categories, months, and regions.

The results show that Electronics leads in sales, while Clothing contributes the highest profit. Loss-making sub-categories and low-profit categories require further investigation to improve profitability. An interactive dashboard with appropriate KPIs, charts, and slicers can help users monitor performance and make informed business decisions.

Before finalizing the report, validate missing order dates, repeated Order IDs, and the relationships between actual sales and monthly category targets.